IS344 Information Retrieval

Homework #2

Due May 10 th 2026 [6 Marks]

Extend the project of the HW#1 To have the following two features:

Write a java code that

1- A web crawler that crawl wikipedia starting from the following this seed URL:

https://en.wikipedia.org/wiki/List_of_pharaohs

2- Build the inverted index for visited pages

3- Get a query (set of a number of words)

4- compute the cosine similarity between each file and the query

5- rank the top k=10 files according to the value of the cosine similarity

Hints:

1- Perform Web Crawling to max pages (N)= 10
2- Build Inverted Index for visited pages + TF
3- Compute IDF for each term in the inverted_index:
    DF ← number of documents containing term
    idf[term] ← log(N / DF)

4- Compute TF-IDF and Document Vectors:
    doc_vectors ← dictionary(doc_id → dictionary(term → weight))

    FOR each term in inverted_index:
        FOR each (doc_id, TF) in inverted_index[term]:
            weight ← TF * idf[term]
            doc_vectors[doc_id][term] ← weight

5- Convert the given query into a numerical vector by calculating TF-IDF for each term in it.
6- Compute Cosine Similarity between the query vector and each document vector.
7- Normalize the similarity scores using the length of docs:
    For each doc:
        do Scores[doc] = Scores[doc]/Length[doc]
8- Rank **Top K** similar Documents.